Q8. Correlation and Covariance

Original Aim: To study how two numerical attributes of the Iris dataset vary together. Calculate the correlation coefficient and covariance between PetalLengthCm and PetalWidthCm. Display both values and state whether the relationship is positive or negative and whether the correlation appears weak or strong.

Aim

Calculate Pearson correlation and sample covariance between PetalLengthCm and PetalWidthCm and interpret direction and strength.

Requirements

Python, Pandas and scikit-learn. Reference: [D8].

Essential theory

Sample covariance measures how centred values vary together, using denominator n-1. Positive covariance means large values tend to accompany large values. Pearson correlation standardizes covariance by the sample standard deviations: r=cov(x,y)/(s_x*s_y). Correlation is unitless and lies between -1 and 1 for nonconstant variables; covariance depends on units and scale.

Step-by-step procedure

1. Select the two petal measurements from matching DataFrame rows.
2. Use corr(method="pearson") for linear association.
3. Use cov(ddof=1) for sample covariance.
4. Display values and interpret sign, magnitude and the limits of the pooled relationship.

Complete Python program

```
from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()
features = [
    "SepalLengthCm", "SepalWidthCm",
    "PetalLengthCm", "PetalWidthCm"
]
species = ["Iris-" + name for name in iris.target_names]
df = pd.DataFrame(iris.data, columns=features)
df["Species"] = pd.Categorical.from_codes(
    iris.target, categories=species
)

x = df["PetalLengthCm"]
y = df["PetalWidthCm"]
correlation = x.corr(y, method="pearson")
covariance = x.cov(y, ddof=1)
print(f"Pearson correlation: {correlation:.6f}")
print(f"Sample covariance (cm^2): {covariance:.6f}")
print("Direction:", "positive" if correlation > 0 else "non-positive")
```

Code explanation

Both Series share row indices, so values are aligned observation by observation. `corr` selects Pearson explicitly; it measures linear association rather than all possible dependence. `cov(ddof=1)` uses the sample convention. For missing values, Pandas generally uses available paired observations; the complete dataset here has 150 pairs. The conditional print displays the direction for this positive result; zero would mean no linear correlation, not necessarily independence.

Verified output

```
Pearson correlation: 0.962865
Sample covariance (cm^2): 1.295609
Direction: positive
```

Observation

Pearson r=0.962865 indicates a strong positive linear association across the pooled 150 flowers: longer petals tend to be wider. The positive covariance 1.295609 cm^2 agrees with the direction, but its magnitude is unit-dependent. Species differences contribute to this aggregate relationship, so it should not be assumed identical within each species. Correlation alone does not establish that one measurement causes the other.

Result

The requested correlation and covariance were calculated from matching rows and interpreted as strong positive pooled association.

Common errors and troubleshooting

- Covariance changes after converting centimetres to millimetres: this is expected; its scale depends on both measurement units.
- A correlation is NaN: check constant inputs, missing pairs or empty/misaligned selections.
- Incorrect pairing after sorting one Series alone: maintain row alignment and do not discard indices independently.

Viva questions and answers

- What is Pearson correlation's range? -1 to +1 when it is defined.
- Is covariance unitless? No; here it has units cm^2.
- Does zero correlation prove independence? No; nonlinear dependence may remain.
- Is a high correlation evidence of causation? No; it is evidence of linear association only.